“Teens Trust AI More Than They Should” A Study of Canadian High School Students’ AI Risk Perception

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December 5, 2025

Project Overview

From June to September 2025, 27 high school students from multiple schools across the Greater Toronto Area (GTA) participated in a social science research project organized by LinkedKey Foundation, and mentored by Professor Tao Wang from School of Information, University of Toronto.

The study is among the first in Canada to explore how teenagers understand and respond to the risks of artificial intelligence (AI) in their daily lives. The target respondents were Grade 9 to 12 students from 16 randomly sampled GTA schools. In total, the project collected 156 valid online survey responses and conducted 32 one-on-one interviews to gain deeper insights into students’ perceptions and behaviors.

AGENDA

1. Background and Research Questions
2. How do students use AI at school
3. What are students’ risk perceptions of AI
4. Conclusions and Implications

Background: A New Generation Born into AI

- Over the past decades, technology has transformed each generation—Gen Z grew up with social media and smartphones, and Gen Alpha with digital learning and smart devices.
- Today, a new generation is emerging as the first to be born into a world where artificial intelligence is fully embedded in daily life, from generative AI to intelligent assistants, recommendation algorithms, and personalized learning tools.
- Unlike earlier “digital natives,” these students interact from childhood with systems that predict, learn, generate content, and intervene, shaping their thinking, learning, and decision-making in unprecedented ways.
- Yet research has not fully captured how they understand, use, or negotiate AI’s influence in their everyday experiences.
- This study addresses that gap by using a mixed approach to explore how AI shapes the lives and learning environments of this emerging AI-born generation.

Research Questions

This study aims to systematically explore the embeddedness and social impact of artificial intelligence in the life of Canadian high school students, with a particular focus on how they coexist with, negotiate with, depend on, or resist AI. The research questions include:

- How do teenagers encounter and use AI in daily life?
- How do teenagers demonstrate understanding and critical awareness of AI systems?
- How do parents and educators influence teenagers’ attitudes toward and usage of AI?

Literature Review: What Are The Potential Risks

Grounded in **risk perception theory** (Slovic, 1987) and aligned with the **OECD Digital Literacy Framework** (OECD, 2023) and **UNESCO's AI Ethics Guidelines** (UNESCO, 2021), prior research identifies four major categories of AI-related risks relevant to youth

1. Performance Risks
AI may encourage over-reliance for schoolwork, which can weaken students' critical thinking, writing ability, and problem-solving skills.

2. Privacy and Cybersecurity Risks
AI may collect extensive personal and behavioral data, which teenagers often underestimate, leaving them vulnerable to data misuse or online threats.

3. Misinformation Risks
AI may generate content that appears authoritative even when inaccurate, exposing teens to "AI illusions" such as fabricated facts, biased outputs, or opaque reasoning processes.

4. Social and Psychological Risks
AI may simulate emotional support or companionship, leading some teens to develop emotional dependence, experience manipulation, or withdraw from real-world social interactions.

Mixed-Method Approach

Survey reveals **what and how much**. Interviews reveal **why and how they feel**. Together, they create a balanced picture of teen AI risk perception.

Quantitative Survey

Conduct online survey towards high school students (Grade 9 to 12)

Randomly selected 16 high schools (12 public and 4 private) across Greater Toronto Area. Only students in these schools can participate in the survey.

Face-to-face Interviews

32 in-person ethnographic interviews plus field observations

Public schools

- Etobicoke Collegiate Institute
- Maple High School
- Harold M. Brathwaite Secondary School
- Newmarket High School
- North Toronto Collegiate Institute
- Craig Kielburger Secondary School
- Rick Hansen Secondary School
- White Oaks Secondary School
- Stephen Lewis Secondary School
- Woburn Collegiate Institute
- Markville Secondary School
- Pickering High School

Private Schools

- Bayview Glen School
- Toronto French School (TFS)
- Holy Trinity School (HTS)
- St. Michael’s College School

Ethical Consideration for Data Collection

Voluntary Participation

All participation was completely voluntary and anonymous with no identifiable data collected

Right to Refuse

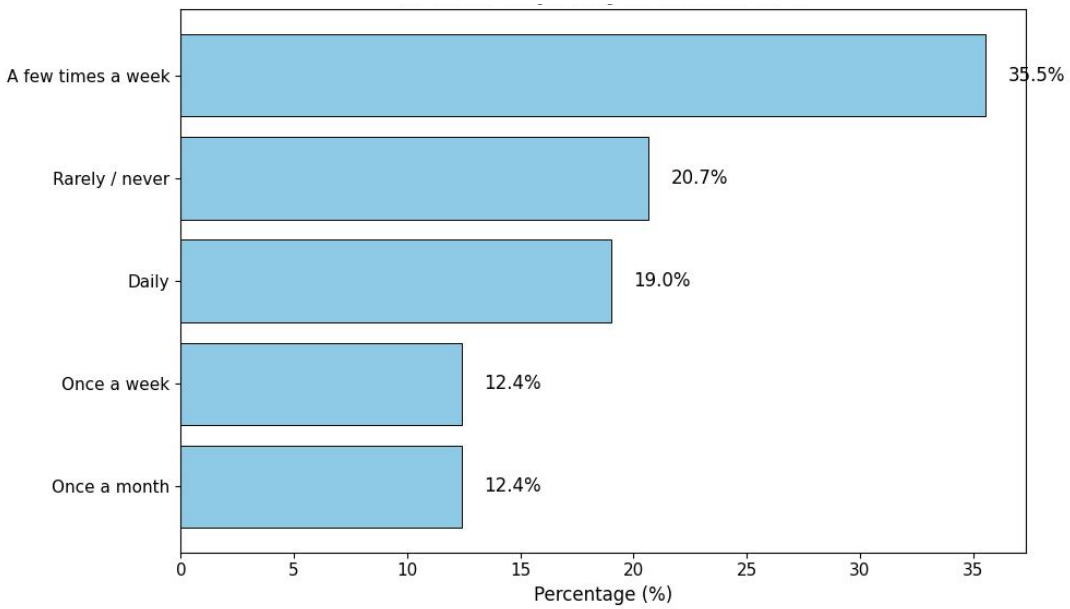
Students could skip or refuse to answer any question that made them feel uncomfortable

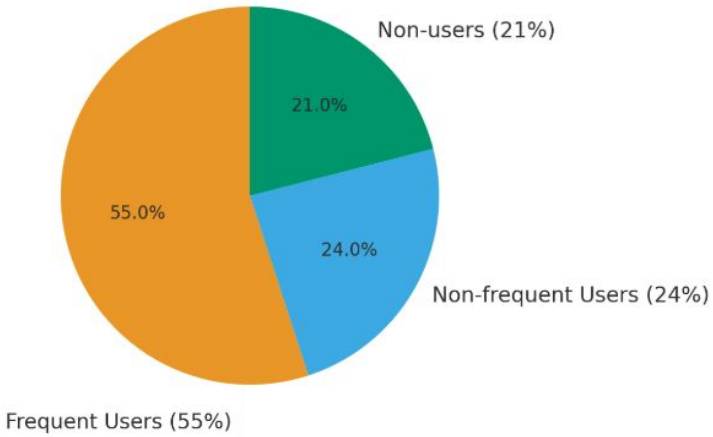
Parental Consent

All consent forms were obtained from parents or guardians before participation

How do student use AI in school?

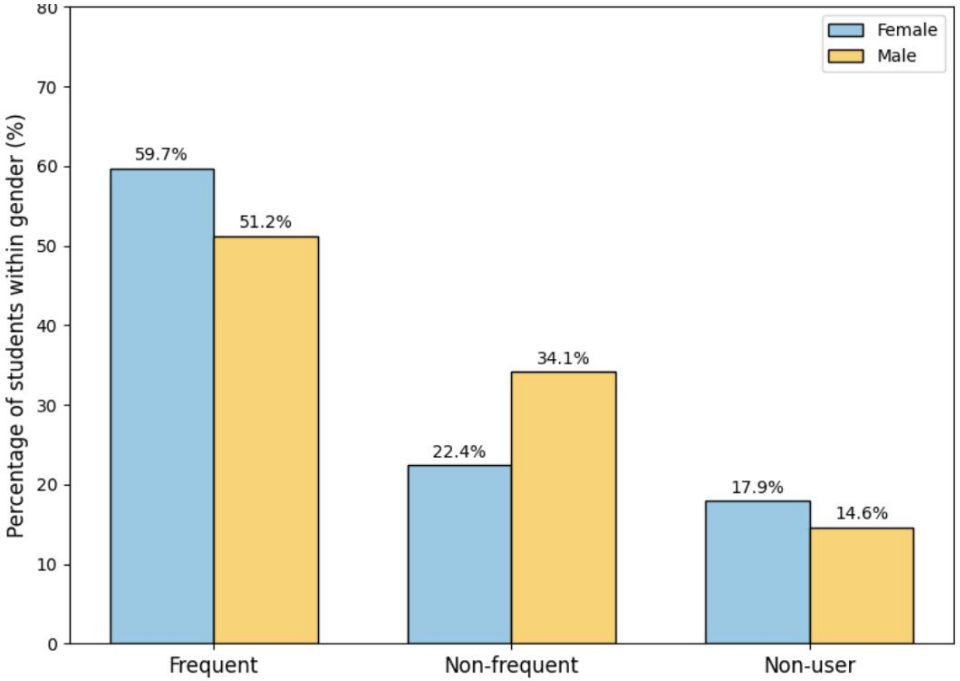
Overall, what percentage of high schoolers use AI in school?





Frequent user: 55% of students use AI more than once a week
Non-frequent user: 24% of students use AI once a week or less
Non-user: 21% of students rarely or never use AI.

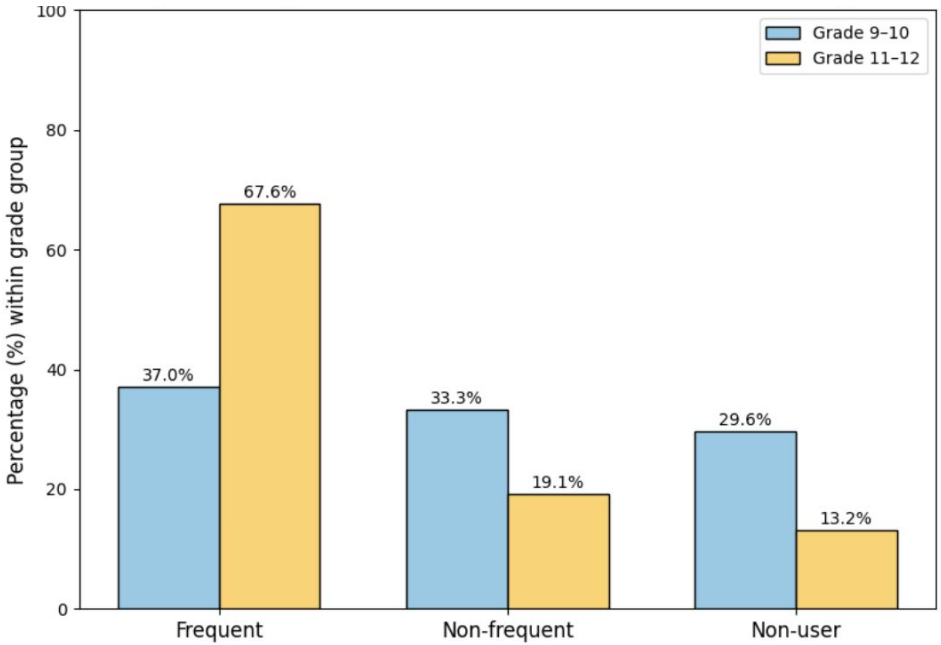
Usage by Female vs Male Students



Female students are more likely to be **frequent AI users** for school, with about 60% using AI daily or a few times a week, compared to 51% of male students.

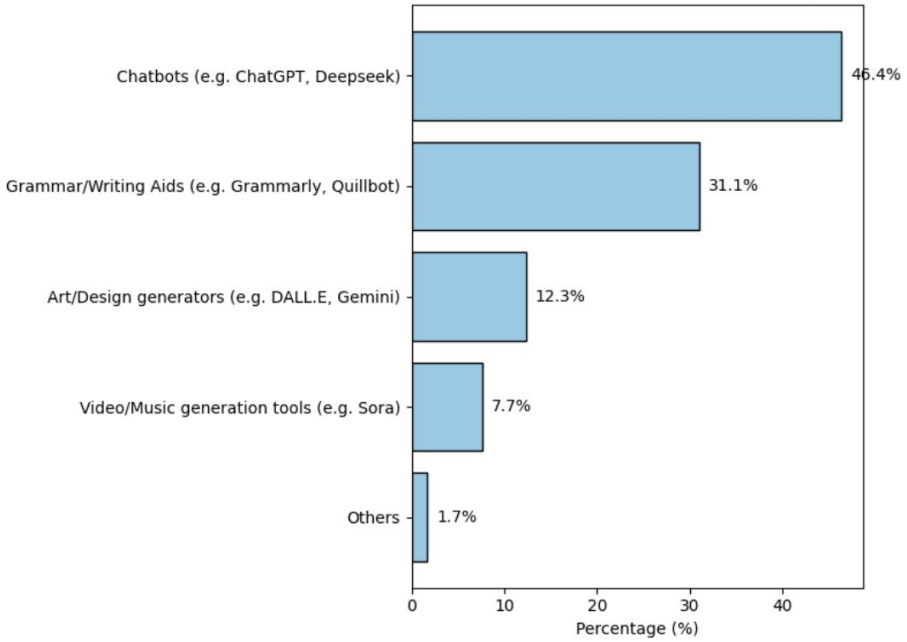
Frequent AI users: who use AI more than once a week at school
Non-frequent AI users: who use AI once a week or less at school
Non-user: who rarely or never use AI

Usage by Senior vs Junior Grade



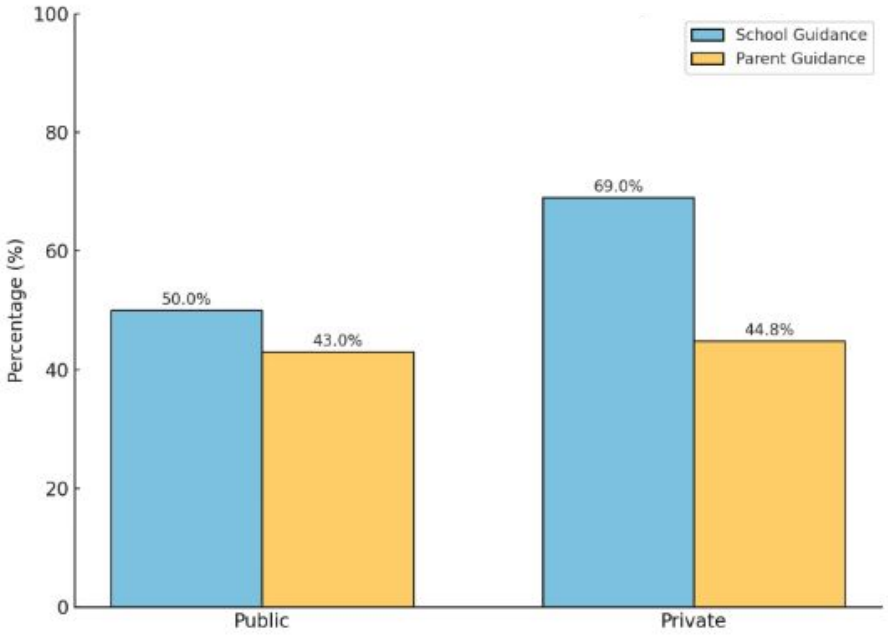
Senior students (Grades 11–12) are much more likely to be **frequent AI users** than junior students in high school.

What do Students Use?



Chatbots and writing aids are two commonly used AI tools.

AI Usage Guidance from School and Parents



Sample size: public school students = 93; private school students = 29

Private school students report receiving AI guidance from their schools at a notably higher rate (69%) than public school students (50%), suggesting stronger institutional emphasis on AI education in private settings.

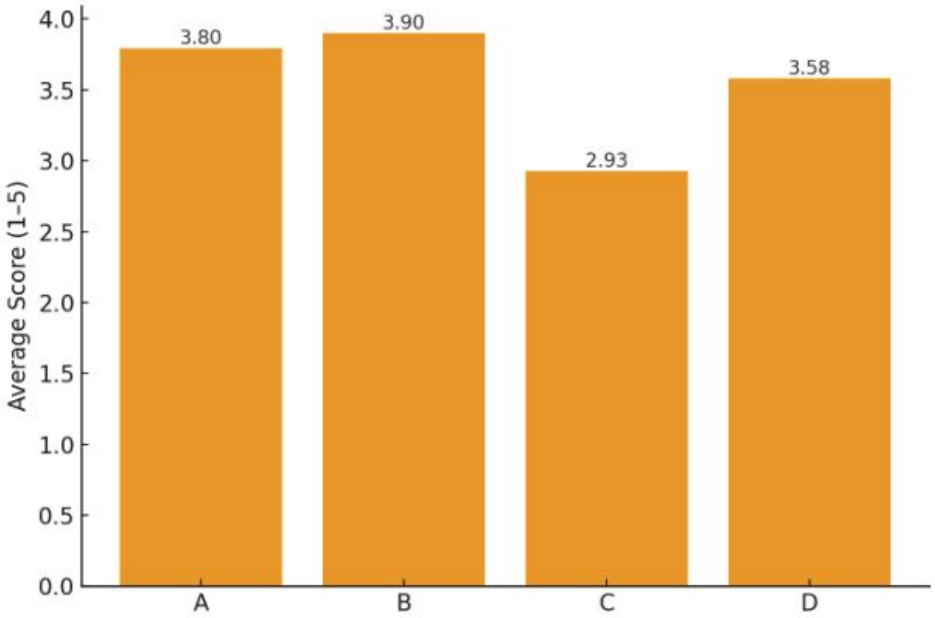
However, parent-provided guidance is similar across both groups,

Findings

- AI is increasingly normalized in schoolwork.
- Female students tend to use AI more frequently than male.
- Senior-grade (11-12) use AI much more frequently than junior-grade (9-10).

What are students’ risk perceptions of AI?

Performance Risks

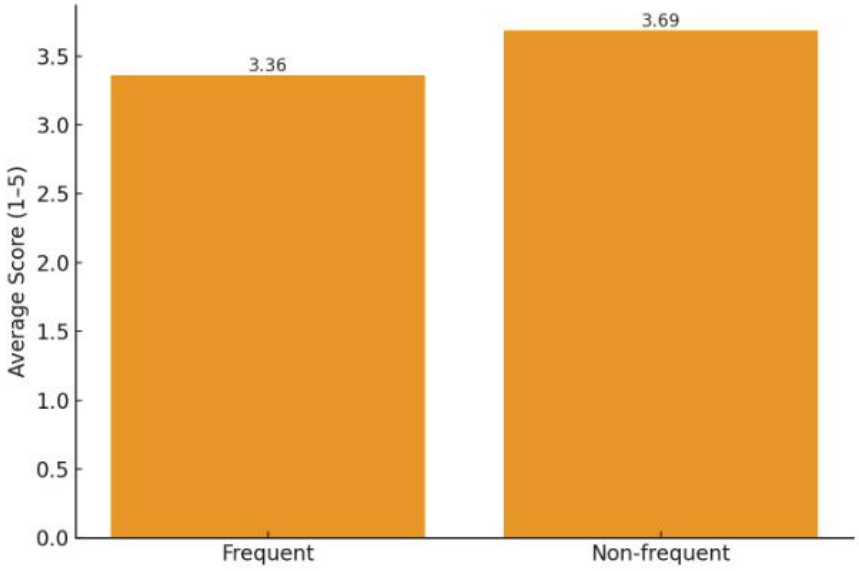


Students show significant worries that using AI will affect their ability to think critically and to learn.

This is a positive sign of their awareness.

The higher the score, the more the students agree with the statement:
A) I believe relying too much on AI for school will harm my own ability to learn
B) I believe relying too much on AI for school will harm my own ability to think critically.
C) I am concerned that some functions of the AI tools I need for school will cause me trouble if it doesn’t work properly
D) I am concerned that regularly using AI for writing help will weaken my own grammar and writing skills over time.

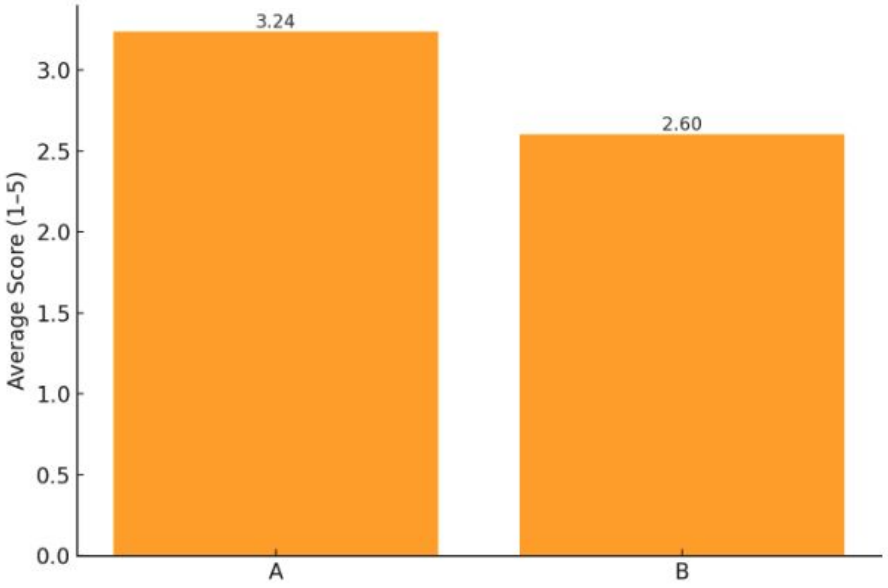
Performance Risks



The frequent AI users are less worried about performance deterioration than non-frequent users.

Frequent AI users: who use AI more than once a week at school
Non-frequent AI users: who use AI once a week or less at school

Privacy and Cybersecurity Risks

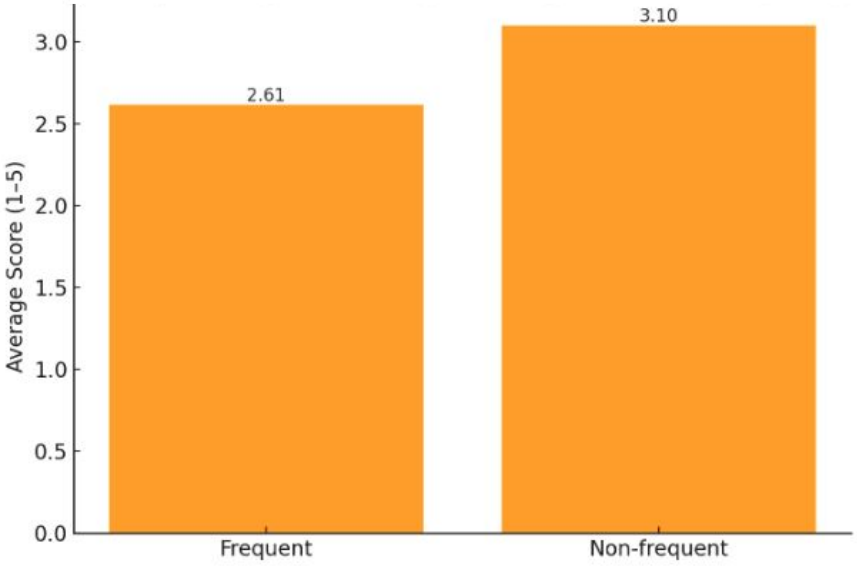


Students show moderate concerns about AI companies misusing personal data.

Low concern about online scams, grooming, or safety threats.

A: I am concerned that AI companies might misuse my personal data or private conversations.
B: I worry that using AI could expose me to online scams, grooming, or other safety threats.

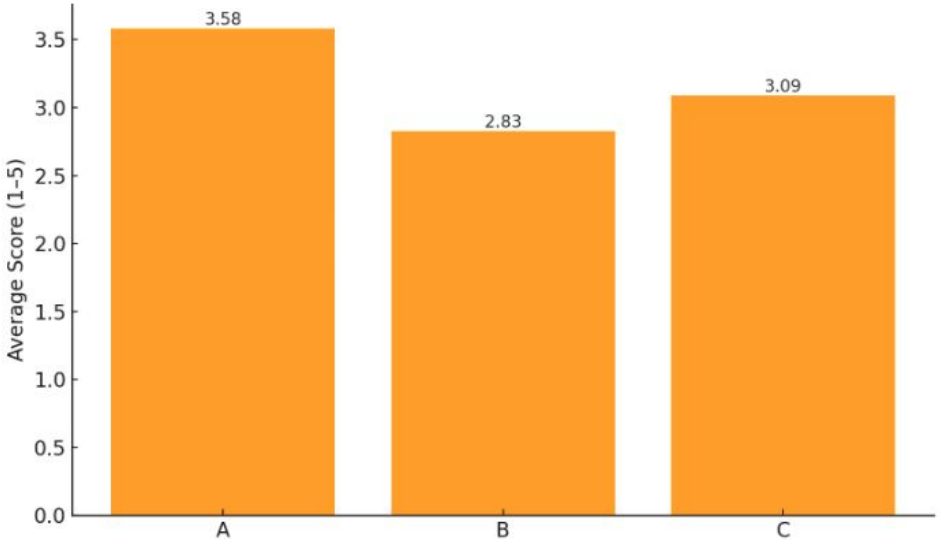
Privacy and Cybersecurity Risks



Non-frequent users show higher concerns than frequent users.

Frequent AI users: who use AI more than once a week at school
Non-frequent AI users: who use AI once a week or less at school

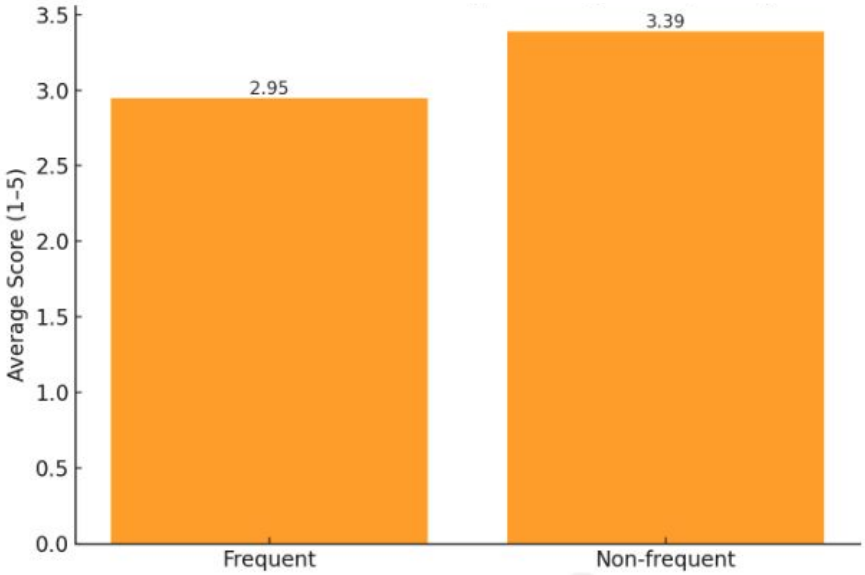
Misinformation Risks



A) I am worried that AI tools will give me incorrect, biased, or harmful information.
B) I am worried that AI tools are so complex that I don't truly understand what they are doing with my data or how they arrive at an answer.
C) I worry about AI making decisions without human supervision.

Teenagers are aware of misinformation as a major issue, while also recognizing—but to a lesser degree—the risks tied to AI's opacity and autonomous decision-making.

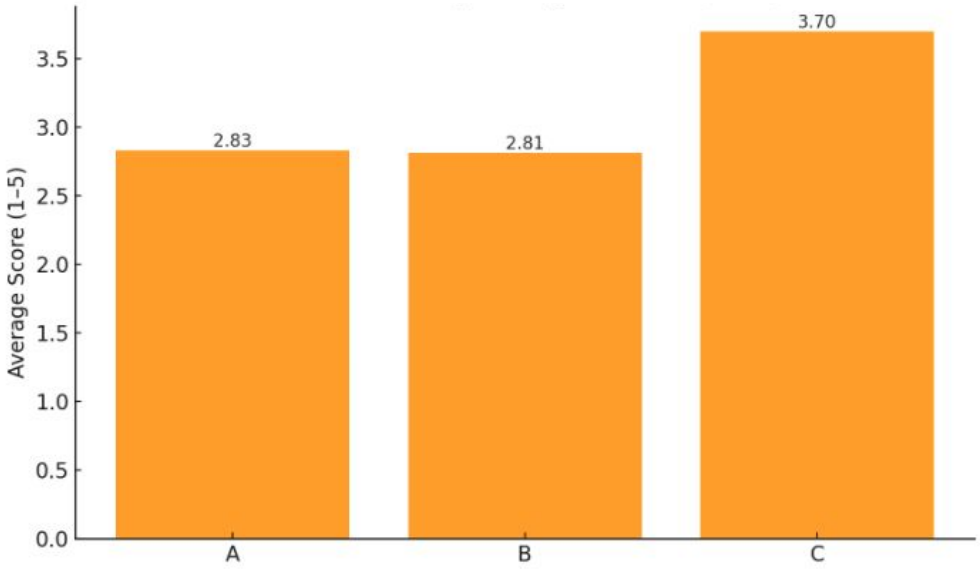
Misinformation Risks



Non-frequent users show higher concerns than frequent users.

Frequent AI users: who use AI more than once a week at school
Non-frequent AI users: who use AI once a week or less at school

Social and Psychological Risks

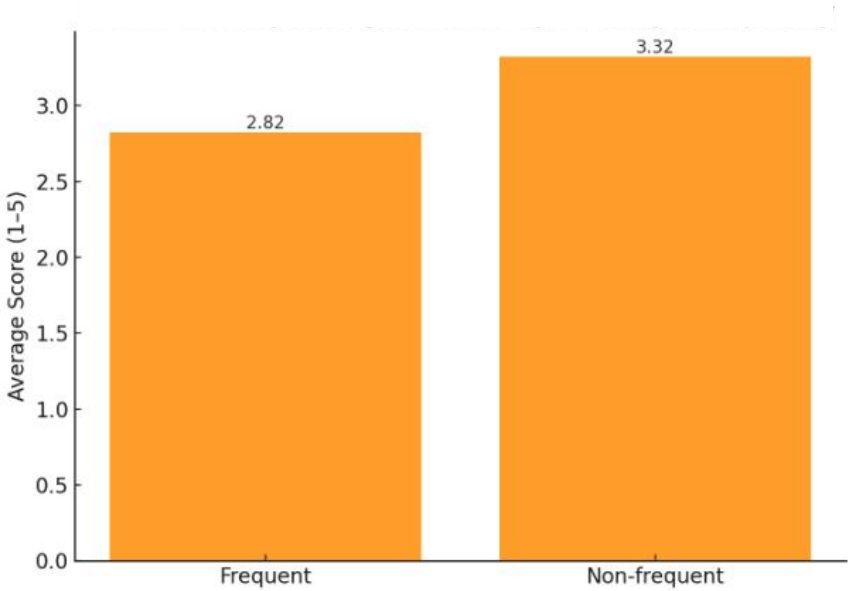


Concern about emotional manipulation (C) received the highest score.

The concerns about social isolation (A) and AI discrimination (B) are moderate, indicating that students view them as not immediate or seriously personally threatening.

A) I worry that using AI chatbots or companions too much could lead to feelings of loneliness or social isolation.
B) I am concerned that AI systems can be unfair or discriminatory, reinforcing negative stereotypes about people.
C) I feel that relying on AI for emotional support or friendship could be emotionally manipulative or unhealthy.

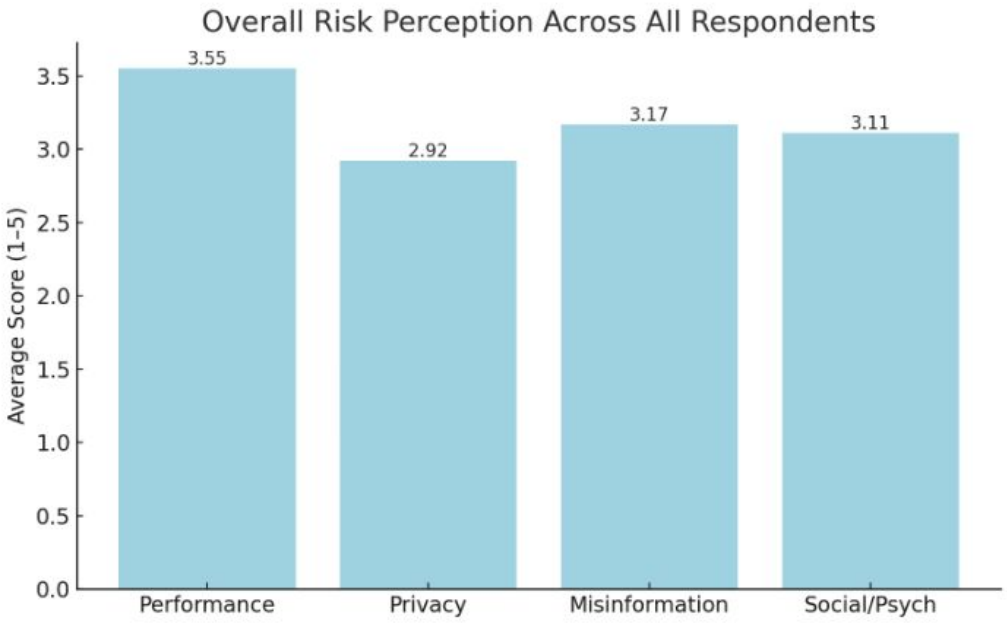
Social and Psychological Risks



Non-frequent users show higher concerns than frequent users.

Frequent AI users: who use AI more than once a week at school
Non-frequent AI users: who use AI once a week or less at school

AI Risk Perception of All Respondents

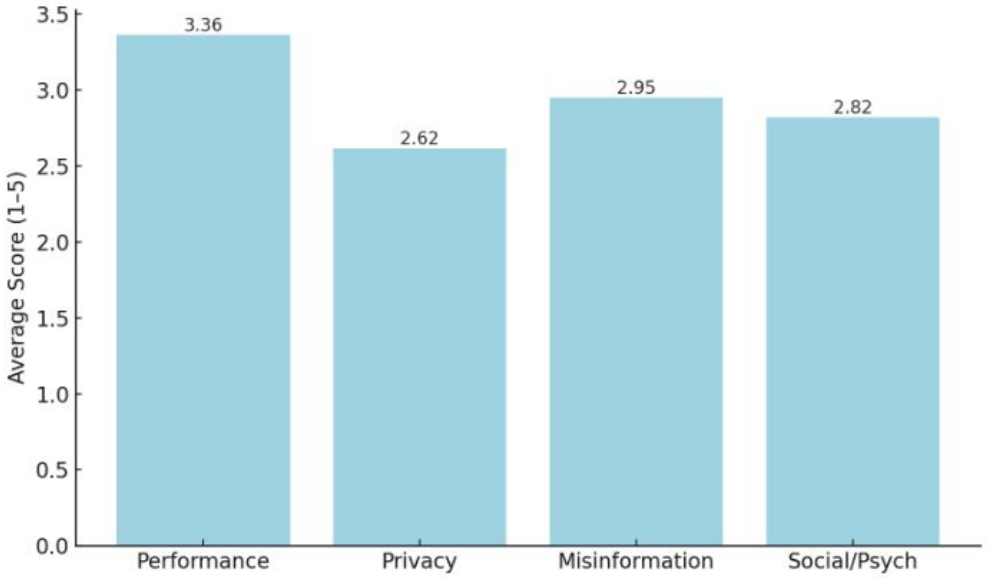


Overall, the risk awareness levels of four types are all low, showing students do not have a strong sense of alertness.

Performance risks are the highest concern, while privacy and cybersecurity risks are the lowest.

1= Strongly disagree, 2 = disagree; 3 = Neutral; 4 = Agree; 5 = Strongly agree

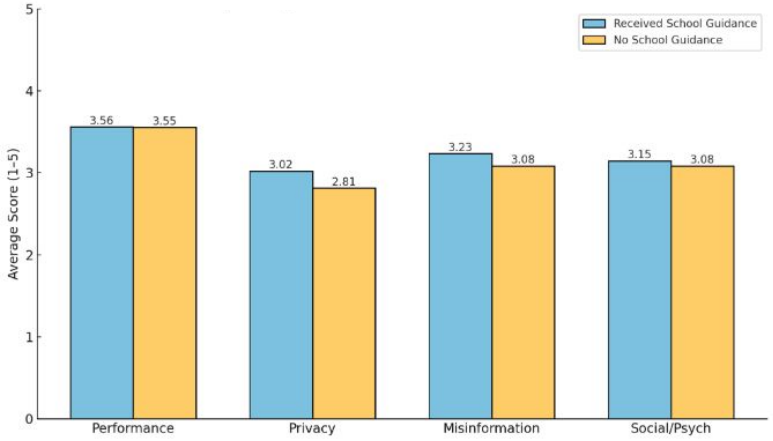
AI Risk Perception of Frequent Users

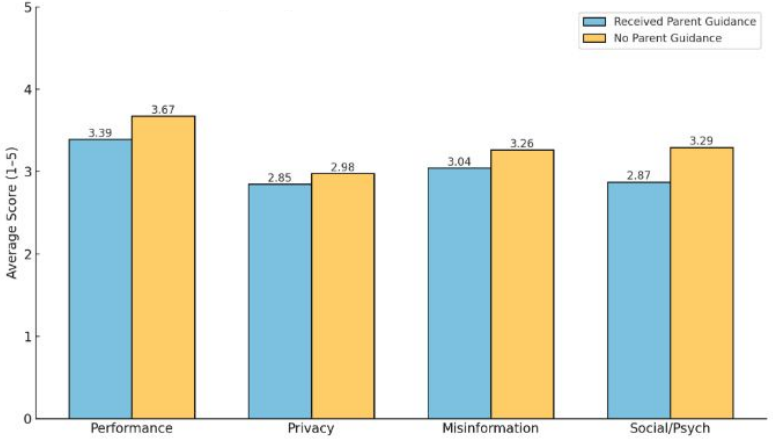


Frequent users have even lower concerns on all four categories of risks.

1= Strongly disagree, 2 = disagree; 3 = Neutral; 4 = Agree; 5 = Strongly agree

AI Risk Perception by Receiving Guidance or Not





Students who received AI guidance from their schools showed **slightly higher awareness of AI risks** across most categories—especially **Privacy** and **Misinformation**. The differences were modest, suggesting that school instruction does not drastically shift students' overall perceptions.

Students who did **not** receive guidance from parents consistently reported **higher concern across all four risk areas**.

Hear What They Said from One-on-One Interviews

Uncertainty and Vulnerability in Personal Data Sharing

“Like basically everything, like, you know, obviously not like my Social Security number, my like passport, not like those type of stuff, but like everything else, sure.”

“How much control do you feel you have over what personal data AI collects about you? 0%.”

Low Concern About Using AI for Emotional Support

“Using Chatgpt for therapy or coping mechanisms and vent thoughts also works.”

“People say don’t do that but I don’t see the harm in it”

“It’s more empathetic and you don’t bother anyone”

“Chatgpt will pretty much align with whatever you say. It is less prominent but still there”.

Some Draw Clear Emotional Boundaries

“Never asks AI for emotional advice- only uses it for academic stuff)”

“Friends would always be better for personal matters”

“AI is not my therapist” (confidently)

Conclusion and Implications

Conclusion

- **AI is deeply integrated into student life**, especially for academic tasks, and usage increases with grade level and among female students.
- **Risk awareness exists but remains low overall**, especially for privacy, cybersecurity, and emotional risks. Students recognize misinformation and performance risks but still rely heavily on AI for convenience.
- **Frequent users show significantly lower concern across all risk categories**, suggesting that habituated use reduces perceived risk even when vulnerabilities remain.
- **A minority of teens use AI for emotional support**, viewing it as empathetic or nonjudgmental. This creates potential psychological risks, though some students maintain clear boundaries.
- **Overall**, the findings reveal a widening gap between actual risks and students' perceived risks, underscoring the need for targeted education.

Implications

For Schools

- Integrate **AI literacy** into curricula, including misinformation detection, ethical use, privacy basics, and AI limitations.
- Provide **clear, age-appropriate guidelines** for academic use to reduce over-reliance and protect learning integrity.
- Implement **structured discussions** about emotional boundaries with AI (e.g., why AI cannot replace human empathy).

For Parents

- Encourage open dialogue about students' AI habits and discuss **risk awareness**, especially around privacy and emotional use.
- Model responsible AI practices at home and align expectations with school guidelines.

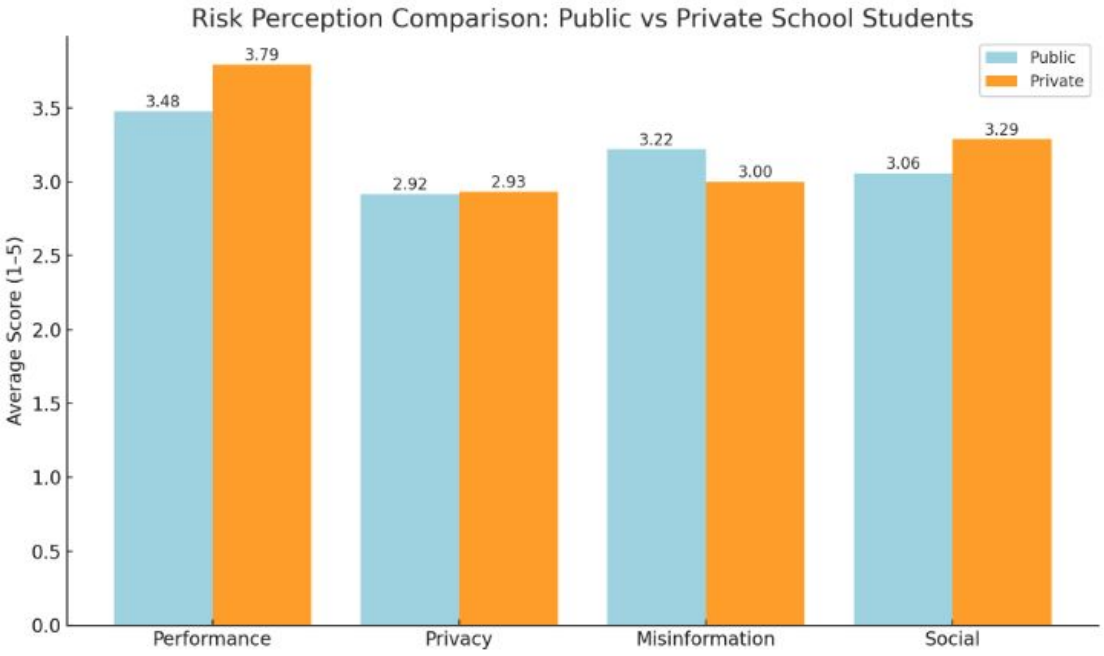
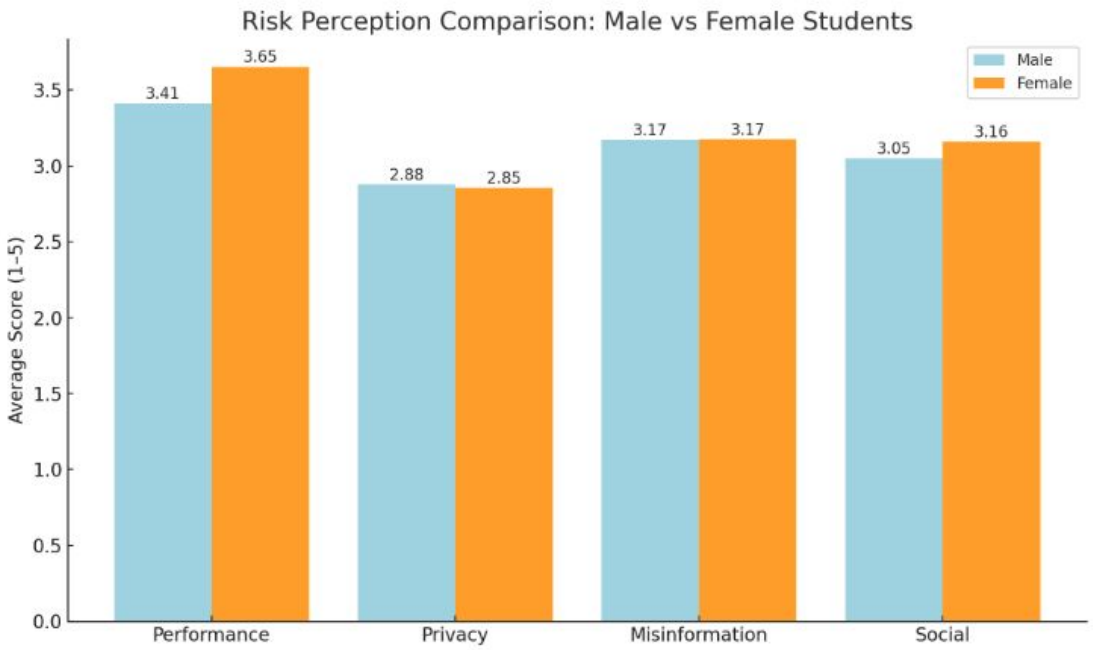
For Policymakers

- Strengthen **AI safety standards** for youth, including data protection and transparency requirements.
- Support teacher training to help educators guide students through safe and effective AI use.
- Promote **nationwide digital literacy initiatives** that empower teens to critically evaluate AI outputs and manage risks.

Thanks!

Backup Slides

AI Risk Perception: Public vs. Private School Students



AI Risk Perception: Male vs. Female Students